PHY 338k Lab Report 4

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1 Introduction

Labs 9–11 of PHY 338k: Electronic Techniques explores

2 Lab 9: Digital Electronics I: Logic Levels and Gates

2.1 Input logic signals and output signal measurement

Include your measurements of the voltage thresholds and a couple of photos of your logical voltage source being read out on the LEDs for different combinations of zeros and ones

2.2 Quad NAND gate

Measure the output of the NAND gate for all four possible input states 00, 01, 10, and 11. Summarize your results with a truth table. Does it agree with what you expect?

Finally, measure and report the actual output voltages of the NAND gate for all four states. You should find a result that is very typical for CMOS logic gates.

Repeat the steps you just did. Do you still get the same truth table, experimentally?

Is the output voltage for the TTL version of the gate the same as the CMOS version, or different?

2.3 NAND gate circuits

2.3.1 NAND implementation of OR gate

Give your measured results in the form of a truth table.

2.3.2 A four input NAND gate circuit

Give your measured results in the form of a truth table.

Include a representative photo of the logic indicator output for one input example.

What logic function does this circuit carry out?

2.4 CMOS logic gates from individual transistors

2.4.1 Inverter with passive pull-up

Include a representative oscilloscope trace showing input and output at a low frequency and at a high frequency.

Give a qualitative description of the circuit behavior.

Also, measure the risetime and fall time of the circuit output and include the result in your report.

2.4.2 Inverter with active pull-up

Include a representative oscilloscope trace showing input and output at a low frequency and at a high frequency.

Give a qualitative description of the circuit behavior.

Also, measure the risetime and fall time of the circuit output and include the result in your report.

What are the main differences between the circuit with active pull-up and the circuit with passive pull- up. Explain why these differences occur. This provides one of the main reasons that essentially all logic circuits use active pull-up.

2.4.3 CMOS NAND gate constructed from individual transistors

Show the circuit diagram your used with just the CD4007 pins and your external connections, and also another diagram showing the complete circuit with the individual transistor connections.

Also show the CD4007 pins on the second diagram.

Show your result for the experimental truth table that you produced with your circuit.

3 Lab 10: Flip-Flops and Sequential Logic

3.1 Flip-flops from NAND gates

3.1.1 Asynchronous SR flip-flop

Verify the functions of the R and S inputs

Report the sequence of R and S values that you used along with the outputs Q and Q-bar. The sequence should include enough steps to fully test the operatoin of the flip-flop including its ability to hold the output for certain changes in R and S.

3.1.2 Clocked SR flip-flop

Report the sequency of R, A, and CLK values you used alon with the outputs Q and Q-bar. Your sequence should include enough steps to fully test the operation of the flip-flop.

3.2 D-type flip-flop

3.2.1 D-type flip-flop operation

Check the operation of the flip-flop by verifying that the D-input appears on the Q-output when and only when clocked.

Assert (pull low) SET and report what happens.

Assert (pull low) RESET and report what happens.

3.2.2 Flip-flop with feedback

Clock the circuit manually and observe the output.

Watch the Clock input and the Q output on an oscilloscope. What is the circuit doing?

Turn the frequency up and measure the propagation delay of the circuit.

Why is this relevant given the feedback connection from Q-bar to D?

3.2.3 Cascaded flip-flops

Using the function generator as the clock input for U1, observe and record the Q output of U2.

3.2.4 Shift register

Using the scope, observe the IN and Q output of the first flip-flop. Why the jitter?

Next, watch the output Q2 along with IN. Repeat for Q3 and Q4. What is the circuit doing?

4 Lab 11: Microcontroller Applications

4.1 Getting started with the Arduino IDE

You should now see the program running your Arduino, i.e. you should see the LED on the board blinking on and off

You should now see the LED blinking with your new time periods.

4.2 Connecting a display

Get the output text on the display

Change the program to output a few of your favorite words and verify that these new words are output on the display.

4.3 Measuring an analog voltage

Verify that the signal is the same as what you see on the oscilloscope

*aliasing - occurs due to periodic sampling of a periodic wave

At about what frequency does the display fail to keep up with the input sine wave?

See if you can get the Arduino to record a faster sine wave. Verify that you can now see a faster sine wave on the display.

What's the fastest sine wave that you can faithfully record now?

4.4 Digital PID control loop

Open a serial plotter window with the baud rate to 9600 and view the resulting voltages

You should now see the voltage drop to zero for a period of time, after which the feedback loop kicks in. What happens as the feedback tries to stabilize the voltage to the set-point?

Using the oscilloscope, observe the direct digital output at I/O pin 6, and also view the filtered output.

comment on how effective the filter is at removing the PWM from the output

Try changing the parameters kp, ki, and kd to see what happens.

Record what happens for some representative set of parameters. What parameters cause the voltage to converge most quickly to the setpoint? Under what conditions is the voltage vs. time more stable? Less stable? Completely unstable?

Conclusion

Labs 9-11

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